



Lecture 9: Introduction to Power System Protection

EEE 321: Power System I

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Presentation Outline

- ❑ Introduction to Protection System
- ❑ Faults in Power Systems
- ❑ Switchgear Equipment- Fuses, Relays, Circuit Breakers
- ❑ Types of Functional Relays- Overcurrent, Differential, Distance Protection

Introduction to Protection System

- ❑ A great demand for electrical energy is a notable feature of modern civilisation. The importance of electric supply in everyday life has reached such a stage that it is desirable to **protect the power system from harm during fault conditions and to ensure maximum continuity of supply**. For this purpose, means must be provided to switch on or off generators, transmission lines, distributors and other equipment under both normal and abnormal conditions. This is achieved by an apparatus called switchgear. A switchgear essentially consists of switching and protecting devices such as switches, fuses, circuit breakers, relays etc.
- ❑ The apparatus used for switching, controlling and protecting the electrical circuits and equipment is known as switchgear.
- ❑ During normal operation, switchgear permits to switch on or off generators, transmission lines, distributors and other electrical equipment. On the other hand, **when a failure (e.g. short circuit) occurs on any part of power system, a heavy current flows through the equipment**, threatening damage to the equipment and interruption of service to the customers. **However, the switchgear detects the fault and disconnects the unhealthy section from the system. In this way, switchgear protects the system from the damage and ensures continuity of supply.**
- ❑ Essential features of switchgear include:
 - (i) Complete reliability
 - (ii) Absolutely certain discrimination
 - (iii) Quick operation
 - (iv) Provision for manual control
 - (v) Provision for instruments

Faults in Power Systems

Most of the failures in power system lead to short-circuit fault and cause heavy current to flow in the system.

Effects of short-circuit. When a short-circuit occurs, the current in the system increases to an abnormally high value while the system voltage decreases to a low value.

(i) The heavy current due to short-circuit causes excessive heating which may result in fire or explosion. Sometimes short-circuit takes the form of an arc and causes considerable damage to the system. For example, an arc on a transmission line not cleared quickly will burn the conductor severely causing it to break, resulting in a long time interruption of the line.

(ii) The low voltage created by the fault has a very harmful effect on the service rendered by the power system. If the voltage remains low for even a few seconds, the consumers' motors may be shut down and generators on the power system may become unstable.

Due to above detrimental effects of short-circuit, it is desirable and necessary to disconnect the faulty section and restore normal voltage and current conditions as quickly as possible.

Calculations of these short-circuit currents are important for the following reasons.

- (i) A short-circuit on the power system is cleared by a circuit breaker or a fuse. It is necessary, therefore, to know the maximum possible values of short-circuit current so that switchgear of suitable rating may be installed to interrupt them.
- (ii) The magnitude of short-circuit current determines the setting and sometimes the types and location of protective system.
- (iii) The magnitude of short-circuit current determines the size of the protective reactors which must be inserted in the system so that the circuit breaker is able to withstand the fault current.
- (iv) The calculation of short-circuit currents enables us to make proper selection of the associated apparatus (*e.g.* bus-bars, current transformers etc.) so that they can withstand the forces that arise due to the occurrence of short circuits.

Faults in Power Systems

A fault occurs when two or more conductors that normally operate with a potential difference come in contact with each other. These faults may be caused by sudden failure of a piece of equipment, accidental damage or short-circuit to overhead lines or by insulation failure resulting from lightning surges. Irrespective of the causes, the faults in a 3-phase system can be classified into two main categories *viz.*

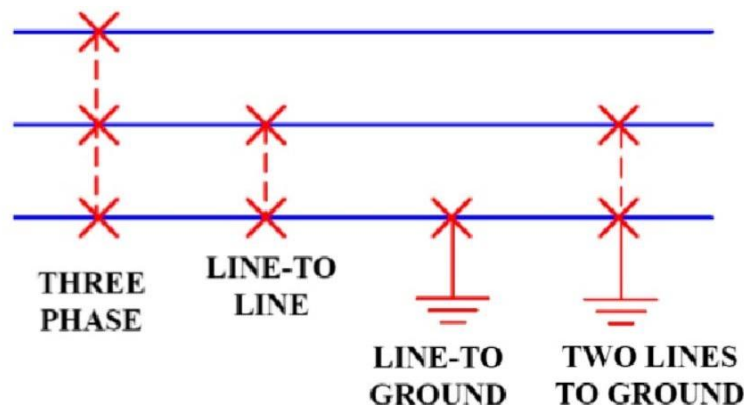
(i) Symmetrical faults (ii) Unsymmetrical faults

(i) **Symmetrical faults.** That fault which gives rise to symmetrical fault currents (*i.e.* equal fault currents with 120° displacement) is called a symmetrical fault. The most common example of symmetrical fault is when all the three conductors of a 3-phase line are brought together simultaneously into a short-circuit condition.

(ii) **Unsymmetrical faults.** Those faults which give rise to unsymmetrical currents (*i.e.* unequal line currents with unequal displacement) are called unsymmetrical faults. The unsymmetrical faults may take one of the following forms :

(a) Single line-to-ground fault (b) Line-to-line fault (c) Double line-to-ground fault

The great majority of faults on the power system are of unsymmetrical nature; the most common type being a short-circuit from one line to ground.



Severity Ranking: The severity of faults, from least to most severe, is generally ranked as follows: LG < LL < LLG < LLL

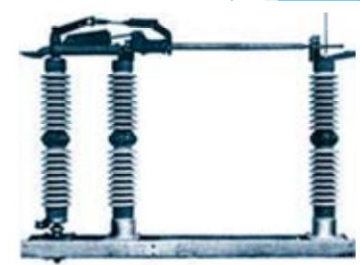
Switchgear Equipment

Switchgear covers a wide range of equipment concerned with switching and interrupting currents under both normal and abnormal conditions. It includes switches, fuses, circuit breakers, relays and other equipment. A brief account of these devices is given below.

1. Switches:

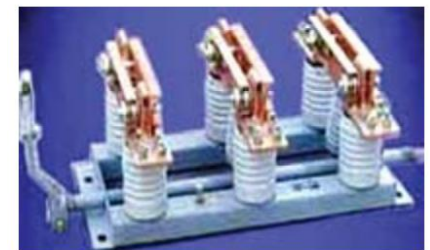
A switch is a device which is used to open or close an electrical circuit in a convenient way. It can be used under full-load or no-load conditions **but it cannot interrupt the fault currents. When the contacts of a switch are opened, an arc is produced in the air between the contacts. This is particularly true for circuits of high voltage and large current capacity.**

Air-break switch: **It is an air switch and is designed to open a circuit under load.** In order to quench the arc that occurs on opening such a switch, special arcing horns are provided. Arcing horns are pieces of metals between which arc is formed during opening operation. As the switch opens, these horns are spread farther and farther apart. Consequently, the arc is lengthened, cooled and interrupted. Air-break switches are generally used outdoor for circuits of medium capacity such as lines supplying an industrial load from a main transmission line or feeder.



Air - break Switch

Isolator or disconnecting switch: **It is essentially a knife switch and is designed to open a circuit under no load.** Its main purpose is to isolate one portion of the circuit from the other and is not intended to be opened while current is flowing in the line. Such switches are generally used on both sides of circuit breakers in order that repairs and replacement of circuit breakers can be made without any danger. They should never be opened until the circuit breaker in the same circuit has been opened and should always be closed before the circuit breaker is closed.



Isolator Switch

Oil switches: As the name implies, **the contacts of such switches are opened under oil, usually transformer oil.** The effect of oil is **to cool and quench the arc** that tends to form when the circuit is opened. **These switches are used for circuits of high voltage and large current carrying capacities.**

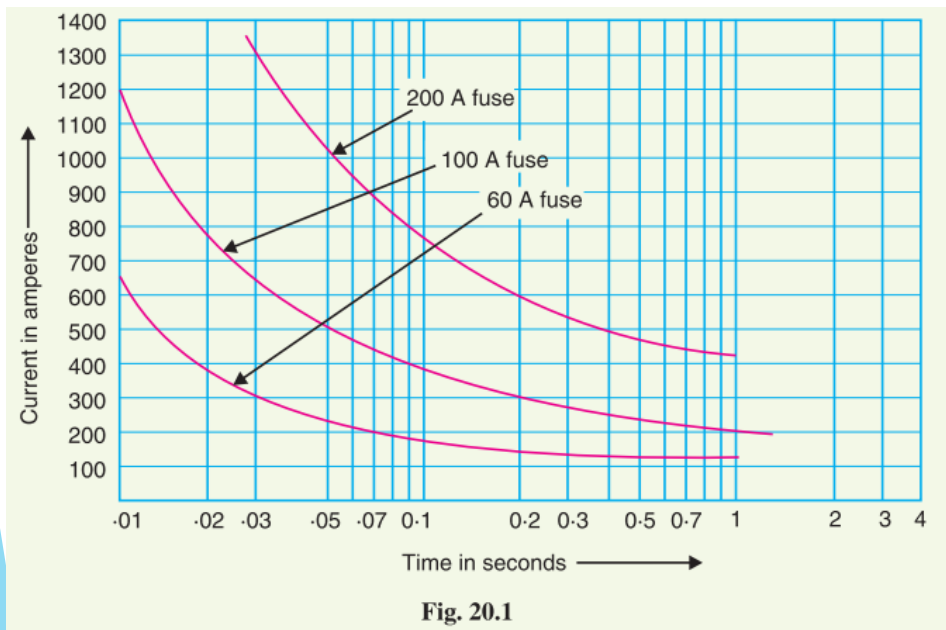


Oil switch

Switchgear Equipment

2. Fuses:

A fuse is a **short piece of wire or thin strip which melts when excessive current flows through it for sufficient time**. It is inserted in series with the circuit to be protected. Under normal operating conditions, the fuse element is at a temperature below its melting point. Therefore, it carries the normal load current without overheating. However, when a short circuit or overload occurs, the current through the fuse element increases beyond its rated capacity. This raises the temperature and the fuse element melts (or blows out), disconnecting the circuit protected by it.



Type of Fuses:

1. Low Voltage Fuses
 - (i) Semi-enclosed rewirable fuse
 - (ii) High-Rupturing Capacity (HRC) fuse
2. High Voltage Fuses

The time required to blow out the fuse depends upon the magnitude of excessive current. **The greater the current, the smaller is the time taken by the fuse to blow out.** In other words, a fuse has **inverse time-current characteristics** as shown in Fig. 20.1. Such a characteristic permits its use for overcurrent protection.

Switchgear Equipment

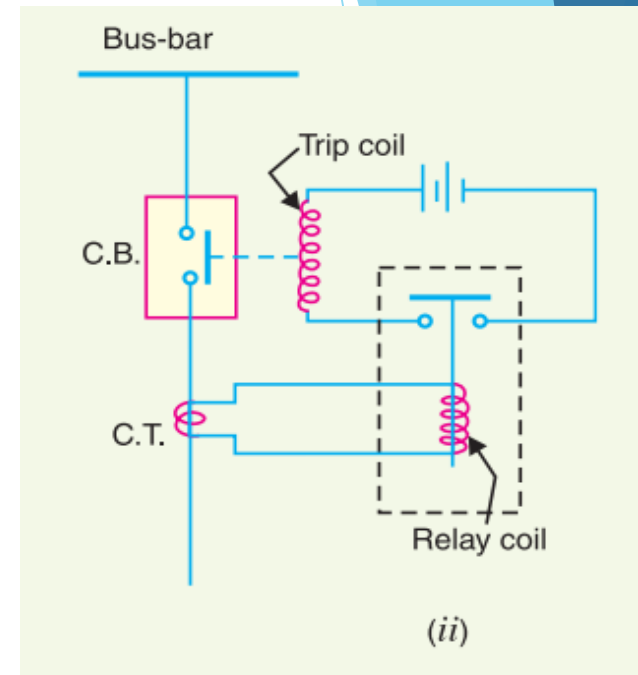
3. Relays:

A relay is a device which detects the fault and supplies information to the breaker for circuit interruption. Fig. 16.1 (ii) shows a typical relay circuit.

It can be divided into three parts viz.

- (i) The primary winding of a *current transformer (C.T.) which is connected in series with the circuit to be protected. The primary winding often consists of the main conductor itself.
- (ii) The second circuit is the secondary winding of C.T. connected to the relay operating coil.
- (iii) The third circuit is the tripping circuit which consists of a source of supply, trip coil of circuit breaker and the relay stationary contacts.

Under normal load conditions, the e.m.f. of the secondary winding of C.T. is small and the current flowing in the relay operating coil is insufficient to close the relay contacts. This keeps the trip coil of the circuit breaker unenergised. Consequently, the contacts of the circuit breaker remain closed and it carries the normal load current. When a fault occurs, a large current flows through the primary of C.T.



Pick-up current. It is the minimum current in the relay coil at which the relay starts to operate. So long as the current in the relay is less than the pick-up value, the relay does not operate and the breaker controlled by it remains in the closed position.

Switchgear Equipment

(i) selectivity

(ii) speed

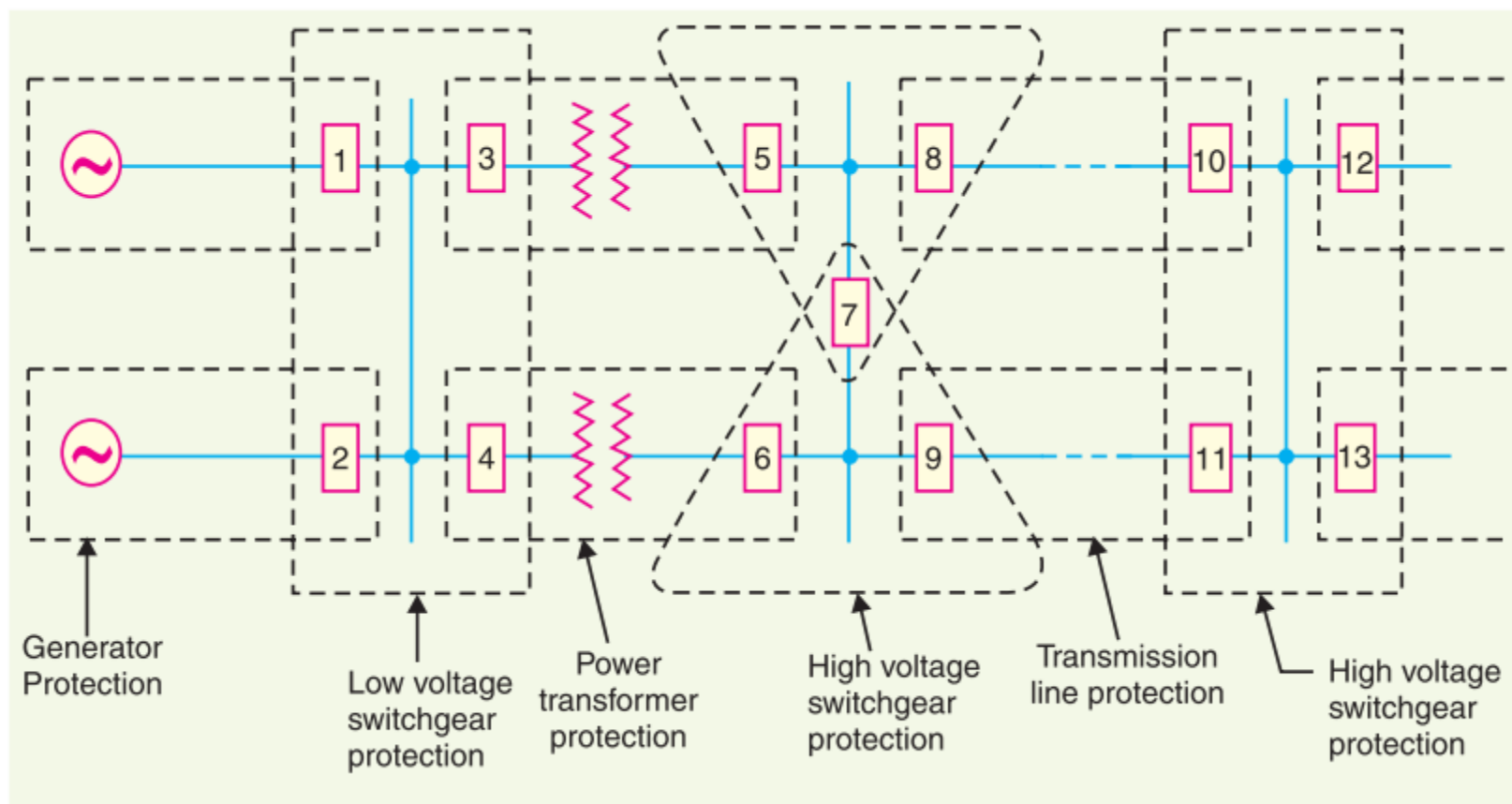
(iii) sensitivity

(iv) reliability

(v) simplicity

(vi) economy

Fundamental requirements of Protective Relaying



In order to provide selectivity to the system, it is a usual practice to divide the entire system into several protection zones. When a fault occurs in a given zone, then only the circuit breakers within that zone will be opened. This will isolate only the faulty circuit or apparatus, leaving the healthy circuits intact.

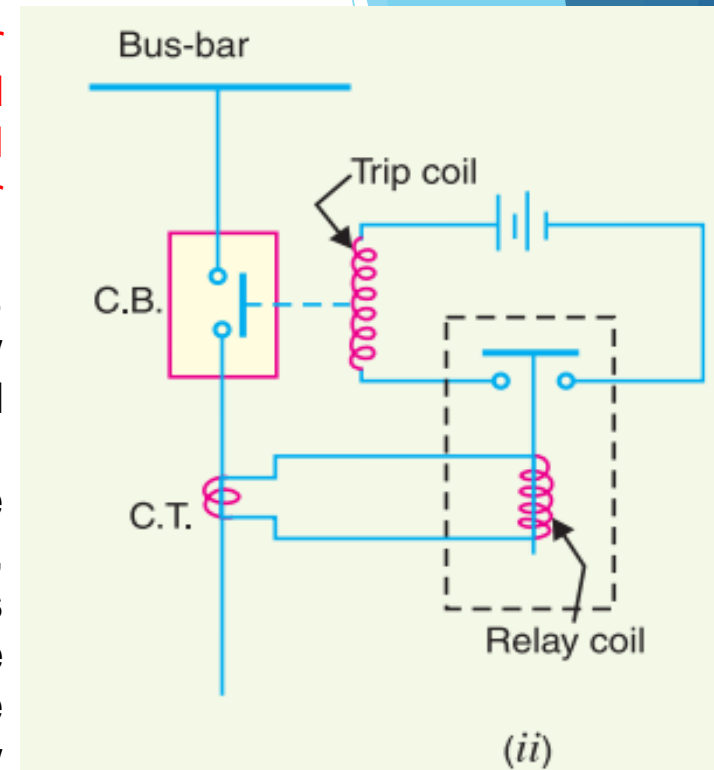
Switchgear Equipment

4. Circuit breakers:

A circuit breaker is an equipment which can open or close a circuit under all conditions viz. no load, full load and fault conditions. It is so designed that it can be operated manually (or by remote control) under normal conditions and automatically under fault conditions. For the latter operation, a relay circuit is used with a circuit breaker.

Fig. 16.1 (i) shows the parts of a typical oil circuit breaker whereas Fig. 16.1 (ii) shows its control by a relay circuit. The circuit breaker essentially consists of moving and fixed contacts enclosed in strong metal tank and immersed in oil, known as transformer oil.

Under normal operating conditions, the contacts remain closed and the circuit breaker carries the full-load current continuously. In this condition, the e.m.f. in the secondary winding of current transformer (C.T.) is insufficient to operate the trip coil of the breaker but the contacts can be opened (and hence the circuit can be opened) by manual or remote control. When a fault occurs, the resulting overcurrent in the C.T. primary winding increases the secondary e.m.f. This energises the trip coil of the breaker and moving contacts are pulled down, thus opening the contacts and hence the circuit. The arc produced during the opening operation is quenched by the oil. It is interesting to note that relay performs the function of detecting a fault whereas the circuit breaker does the actual circuit interruption.



Switchgear Equipment

4. Circuit breakers:

When the contacts of a CB are separated under fault conditions, an arc is produced between them.

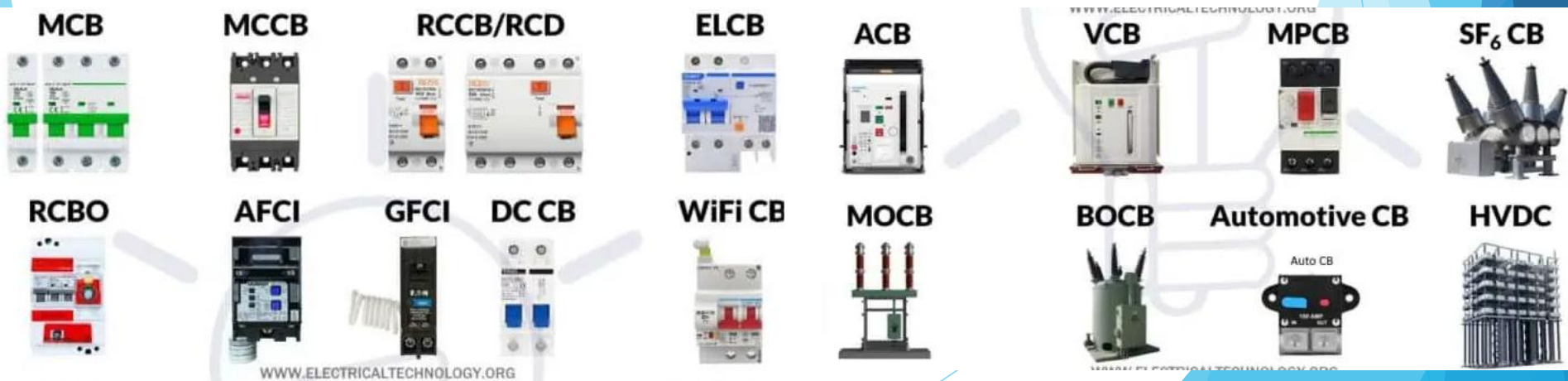
Based on the medium used for arc extinction, CB may be classified into:

(i) Oil CB (ii) Air-blast CB (iii) S₆ CB (iv) Vacuum CB

It is primarily rated as breaking current that it can carry before opening the moving contact.

- (i) *Oil circuit breakers* which employ some insulating oil (e.g., transformer oil) for arc extinction.
- (ii) *Air-blast circuit breakers* in which high pressure air-blast is used for extinguishing the arc.
- (iii) *Sulphur hexafluoride circuit breakers* in which sulphur hexafluoride (SF₆) gas is used for arc extinction.
- (iv) *Vacuum circuit breakers* in which vacuum is used for arc extinction.

Each type of circuit breaker has its own advantages and disadvantages. In the following sections, we shall discuss the construction and working of these circuit breakers with special emphasis on the way the arc extinction is facilitated.



Switchgear Equipment

4. Circuit breakers:

Sulphur Hexafluoride (SF₆) CB

In such circuit breakers, sulphur hexafluoride (SF₆) gas is used as the arc quenching medium. The SF₆ is an electro-negative gas and has a strong tendency to absorb free electrons. The contacts of the breaker are opened in a high pressure flow of SF₆ gas and an arc is struck between them. The conducting free electrons in the arc are rapidly captured by the gas to form relatively immobile negative ions. This loss of conducting electrons in the arc quickly builds up enough insulation strength to extinguish the arc. The SF₆ circuit breakers have been found to be very effective for high power and high voltage service.

Advantages. Due to the superior arc quenching properties of SF₆ gas, the SF₆ circuit breakers have many advantages over oil or air circuit breakers. Some of them are listed below :

- ➡ (i) Due to the superior arc quenching property of SF₆, such circuit breakers have very short arcing time.
- ➡ (ii) Since the dielectric strength of SF₆ gas is 2 to 3 times that of air, such breakers can interrupt much larger currents.
- (iii) The SF₆ circuit breaker gives noiseless operation due to its closed gas circuit and no exhaust to atmosphere unlike the air blast circuit breaker.
- (iv) The closed gas enclosure keeps the interior dry so that there is no moisture problem.
- ➡ (v) There is no risk of fire in such breakers because SF₆ gas is non-inflammable.
- (vi) There are no carbon deposits so that tracking and insulation problems are eliminated.
- ➡ (vii) The SF₆ breakers have low maintenance cost, light foundation requirements and minimum auxiliary equipment.

Switchgear Equipment

4. Circuit breakers:



Sulphur Hexafluoride (SF₆) CB

Disadvantages

- ➔ (i) SF₆ breakers are costly due to the high cost of SF₆.
- (ii) Since SF₆ gas has to be reconditioned after every operation of the breaker, additional equipment is required for this purpose.

Applications. A typical SF₆ circuit breaker consists of interrupter units each capable of dealing with currents upto 60 kA and voltages in the range of 50—80 kV. A number of units are connected in series according to the system voltage. SF₆ circuit breakers have been developed for voltages 115 kV to 230 kV, power ratings 10 MVA to 20 MVA and interrupting time less than 3 cycles.

Switchgear Equipment

Difference between Fuse and Circuit Breaker:

It is worthwhile to indicate the salient differences between a fuse and a circuit breaker in the tabular form.

S. No.	Particular	Fuse	Circuit breaker
1.	<i>Function</i>	It performs both detection and interruption functions.	It performs interruption function only. The detection of fault is made by relay system.
2.	<i>Operation</i>	Inherently completely automatic.	Requires elaborate equipment (<i>i.e.</i> relays) for automatic action.
3.	<i>Breaking capacity</i>	Small	Very large
4.	<i>Operating time</i>	Very small (0.002 sec or so)	Comparatively large (0.1 to 0.2 sec)
5.	<i>Replacement</i>	Requires replacement after every operation.	No replacement after operation.

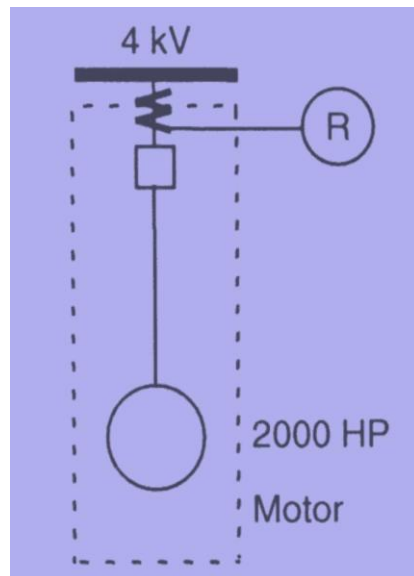
Types of Functional Relays

Overcurrent Protection

Consider the overcurrent protection scheme of the motor shown below.

The rated current for the 2000 HP, 4 kV motor is

$$I_{rated} = \frac{2000 \times 746}{\sqrt{3} \times 4000} = 215 A$$



Allow 25% over load= 54 Amp Add 20% safety margin=43 Amp Total = 312 A

- Any current above set value (chosen to be 312 A) may be taken to mean that a fault or some abnormal condition exists in the zone of protection of the motor.
- The relay should be designed to operate and trip the circuit breaker above this setting.
- If desired, the relay may be connected to sound an alarm so that operator can intervene, and trip the CB manually, or take some other action

Types of Functional Relays

Overcurrent Protection

The level above which the relay operates is known as the **pick-up** setting of the relay. For currents above this value the relay operates. The relay takes no action if the current is smaller than the pick-up.

The operating characteristic of an over-current relay is plot of operating time vs. relay current given below.

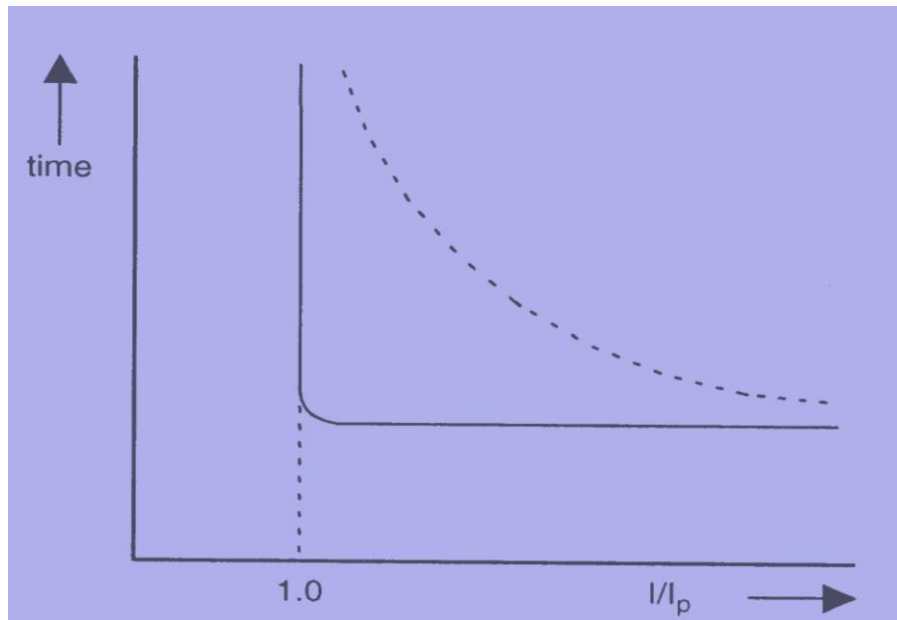


Fig. Characteristic of an overcurrent relay

- The current is normalized (actual current/pickup setting)
- Actual time of operation will depend on type of relay
- The ideal level detector would have the solid line characteristic

Applications of inverse time-delay overcurrent protection

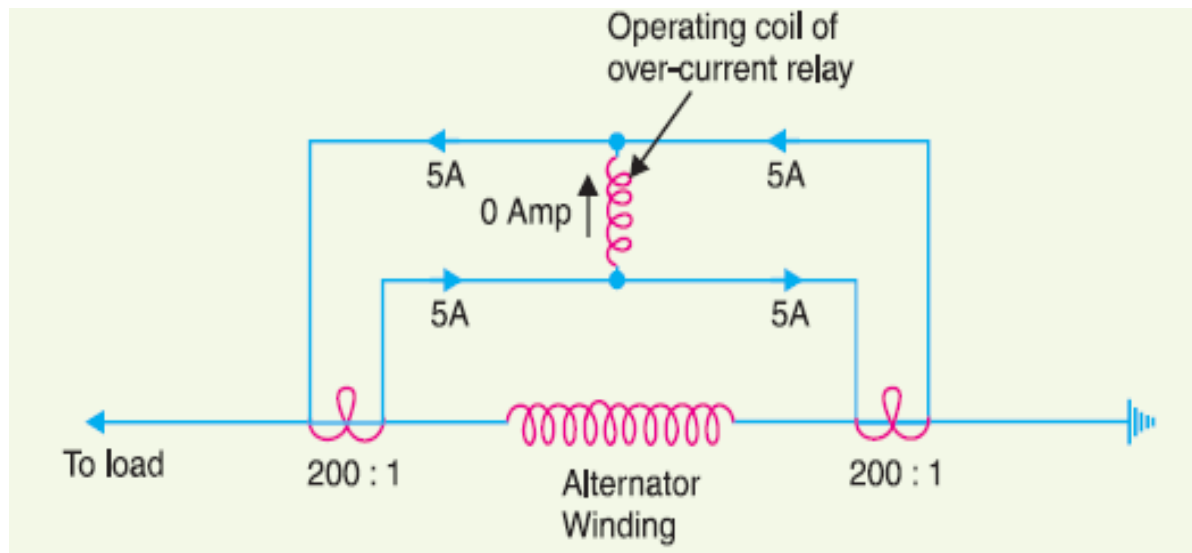
- Principal application of over-current relay is in a radial system where they provide both phase and ground protection
- Overcurrent relays are also used in industrial systems and on sub-transmission lines which cannot justify more expensive protection such as distance or pilot relays

Types of Functional Relays

Differential Protection

A **differential relay** is one that operates when the difference of two or more similar electrical quantities exceeds a pre-determined value.

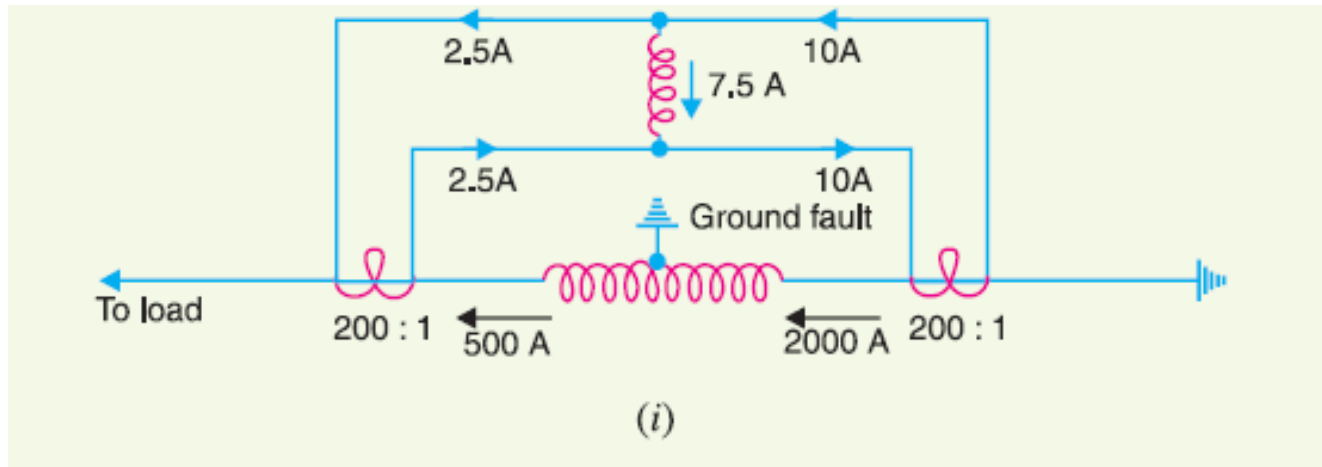
(i) Consider the arrangement of an overcurrent relay connected to operate as a differential relay in the following figure. Under normal operating conditions, suppose the alternator winding carries a normal current of 1000 A. Then the currents in the two secondaries of CT's are equal. These currents will merely circulate between the two CT's and no current will flow through the differential relay. Therefore, the relay remains inoperative.



Types of Functional Relays

Differential Protection

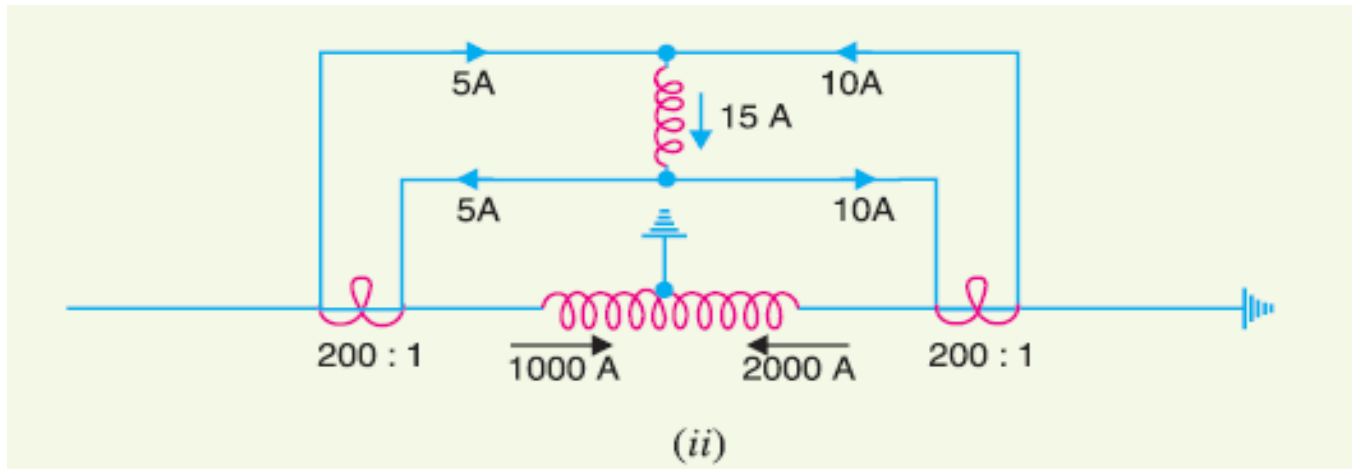
(ii) If a ground fault occurs on the alternator winding as shown in Fig (i), and if some small current (500 A in this case) flows out of one side while a larger current (2000 A) enters the other side then the difference of the CT secondary currents *i.e.* $10 - 2.5 = 7.5$ A will flow through the relay.



Types of Functional Relays

Differential Protection

(iii) If current flows to the fault from both sides as shown in Fig. (ii), then sum of CT secondary currents i.e. $10 + 5 = 15 \text{ A}$ will flow through the relay.



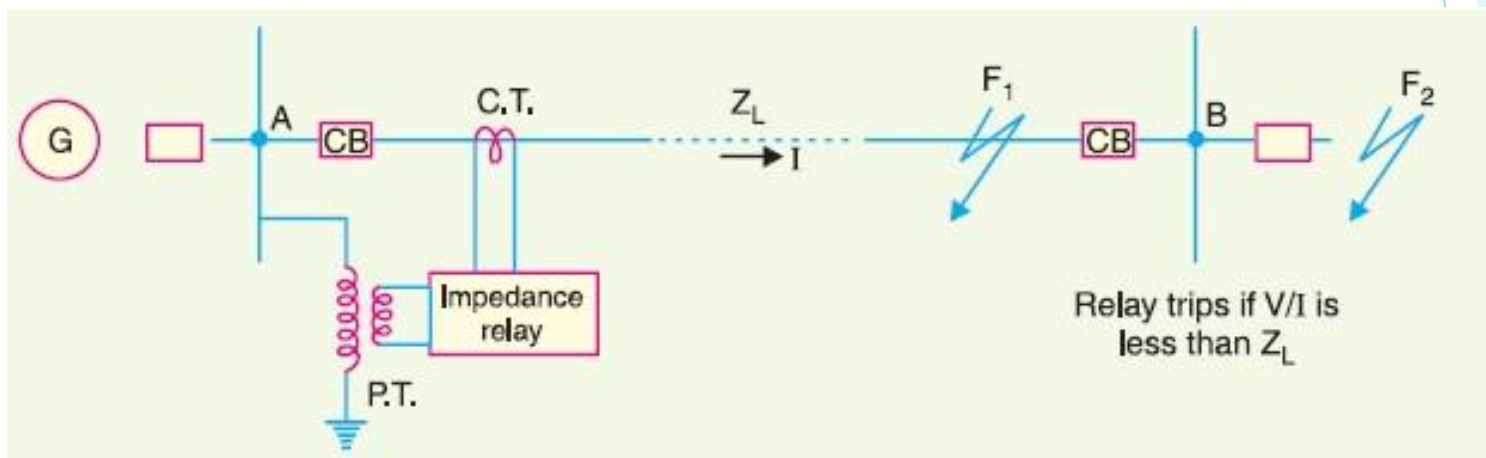
In general, a differential protection is capable of detecting very small magnitudes of fault current. Its only drawback is that it requires currents from the extremities of the zone of protection. This limits its use to,

- Generators
- Transformers
- Motors
- Buses
- Capacitors
- Reactors, etc

Types of Functional Relays

Distance or Impedance Protection

The fig. below illustrates the basic principle of operation of an impedance relay. The voltage element of the relay is excited through a potential transformer (P.T.) from the line to be protected. The current element of the relay is excited from a current transformer (C.T.) in series with the line. The portion AB of the line is the protected zone. Under normal operating conditions, the impedance of the protected zone is Z_L . The relay is so designed that it closes its contacts whenever impedance of the protected section falls below the pre-determined value *i.e.* Z_L in this case.



Now suppose a fault occurs at point F_1 in the protected zone. The impedance $Z (= V/I)$ between the point where the relay is installed and the point of fault will be less than Z_L and hence the relay operates. Should the fault occur beyond the protected zone (say point F_2), the impedance Z will be greater than Z_L and the relay does not operate.

Types of Functional Relays

Distance or Impedance Protection

The principle and operation of Distance Protection relays have already been discussed here. We shall now consider its application for the protection of transmission lines. Fig. 23.12 (i) shows a simple system consisting of lines in series such that power can flow only from left to right.

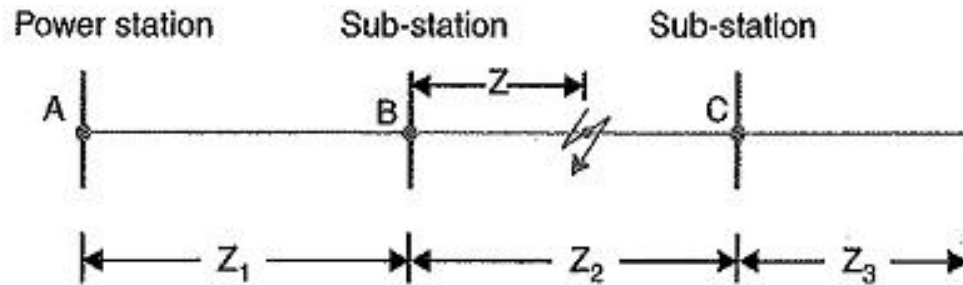


Fig. 23.12 (i)

The relays at A, B and C are set to operate for impedance less than Z_1 , Z_2 and Z_3 respectively. Suppose a fault occurs between sub-stations B and C, the fault impedance at power station and sub-station A and B will be $Z_1 + Z$ and Z respectively. It is clear that for the portion shown, only relay at B will operate. Similarly, if a fault occurs within section AB, then only relay at A will operate. In this manner, instantaneous protection can be obtained for all conditions of operation.

Reference:

'Principles of Power System', V.K. Mehta and Rohit Mehta



QUESTIONS AND DISCUSSIONS

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